Electronic Configuration

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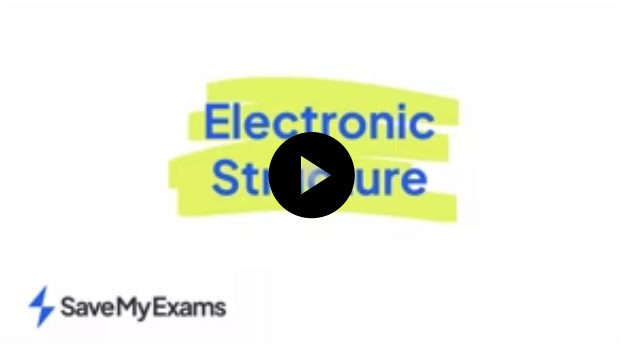

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What is electronic configuration?

Electronic configuration

- We can represent the structure of the atom in two ways: using diagrams called **electron shell diagrams** or by writing out a special notation called the **electronic configuration** (or electronic structure or electron distribution)

Electron shell diagrams

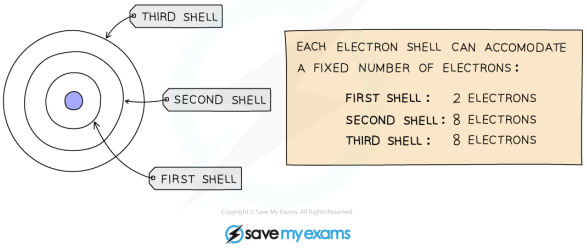
- Electrons orbit the nucleus in **shells** (or **energy levels**) and each shell has a different amount of energy associated with it
- The further away from the nucleus, the more energy a shell has
- Electrons fill the shell closest to the nucleus
- When a shell becomes full of electrons, additional electrons have to be added to the next shell
  - The first shell can hold 2 electrons
  - The second shell can hold 8 electrons
  - The third shell can hold 8 electrons
  - Then the fourth shell starts to fill

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

The third shell can actually hold up to 18 electrons but a simplified 8 electron model is used at this level

- The outermost shell of an atom is called the **valence** shell and an atom is much more stable if it can manage to completely fill this shell with electrons



A simplified model showing the electron shells

- The arrangement of electrons in shells can also be explained using numbers
- Instead of drawing electron shell diagrams, the number of electrons in each electron shell can be written down, separated by commas
- This notation is called the electronic configuration (or electronic structure)
  - E.g. Carbon has 6 electrons, 2 in the 1st shell and 4 in the 2nd shell
    - Its electronic configuration is 2,4

- Electronic configurations can also be written for ions
  - E.g. A sodium atom has 11 electrons, a sodium ion has lost one electron, therefore has 10 electrons; 2 in the first shell and 8 in the 2nd shell
    - Its electronic configuration is 2,8

The electronic configuration of the first twenty elements

Element	Atomic Number	Electronic Configuration
hydrogen	1	1
helium	2	2
lithium	3	2,1
beryllium	4	2,2
boron	5	2,3
carbon	6	2,4

oxygen	8	2,6
fluorine	9	2,7
neon	10	2,8
sodium	11	2,8,1
magnesium	12	2,8,2
aluminium	13	2,8,3
silicon	14	2,8,4
phosphorus	15	2,8,5
sulfur	16	2,8,6
chlorine	17	2,8,7
argon	18	2,8,8
potassium	19	2,8,8,1
calcium	20	2,8,8,2

Note: although the third shell can hold up to 18 electrons, the filling of the shells follows a more complicated pattern after potassium and calcium. For these two elements, the third shell holds 8 and the remaining electrons (for reasons of stability) occupy the fourth shell first before filling the third shell.

Worked Example

Draw and write the electronic structure of magnesium.

Answer:

- Magnesium has 12 electrons in total.
- A maximum of two can fit in the first shell and eight in the second shell.
- The remaining two will occupy the third shell.

Examiner Tips and Tricks

You need to be able to write the electronic configuration of the first twenty elements and their ions. You may see electronic configurations using full stops or '+' signs instead of commas. You would not be penalised for using full stops.

Electron shells & the Periodic Table

How does the electronic structure of an element relate to its location in the Periodic Table?

- There is a clear **relationship** between the electronic configuration and how the Periodic Table is designed
- The number of notations in the electronic configuration will show the number of occupied shells of electrons the atom has, showing the **period** in which that element is in
- The last notation shows the number of outer electrons the atom has, showing the **group** that element is in (for elements in Groups I to VII)
- Elements in the **same group** have the **same** number of outer **shell electrons**

Diagram showing the relationship between the electronic configurations

The electronic configuration for chlorine

- **Period:** The red numbers at the bottom show the number of notations
  - The number of notations is 3
  - Therefore chlorine has 3 occupied shells
- **Group:** The last notation, in this case 7
  - This means that chlorine has 7 electrons in its outer shell
  - Chlorine is therefore in Group 7

The position of chlorine on the Periodic Table

The noble gases

- In most atoms, the outermost shell is not full
- These atoms react with other atoms in order to achieve a full outer shell of electrons making them more stable
- In some cases, atoms lose electrons to entirely empty this shell so that the next shell below becomes a (full) outer shell
  - They then have the electronic structure of a noble gas (Group VIII)
- All of the noble gases are **unreactive** as they have full outer shells and are thus very stable

The Periodic Table showing the location of the noble gases

The noble gases are on the Periodic Table in Group 8/0

Examiner Tips and Tricks

The electrons in the outer shell are also known as valency electrons.

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